Abhijeet Anand, PhD

Independent Fellow, Inter-University Centre for Astronomy and Astrophysics (IUCAA)

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Computational Physicist and Data Scientist with 7+ years of experience developing, maintaining, and improving open-source scientific software and large-scale analysis pipelines for astronomy. Strong background in scientific Python ecosystem, including statistical validation, workflow automation, collaborative Git/GitHub based software development, automated testing and CI/CD workflows, documentation, versioned software releases. Experience in communicating technical results across multidisciplinary teams.

Work Experience

Independent Postdoctoral Fellow *IUCAA, Pune* *Dec 2025 – Present*
Affiliate Scientist *Lawrence Berkeley National Laboratory, USA* *Feb 2026 – Present*

- Lead end-to-end data analysis projects, from preparing noisy datasets and building detection models to validating results and delivering community data products.
- Develop and maintain Python pipelines for one-dimensional sequential data, including transformation, signal processing, classification, and uncertainty analysis.
- Apply supervised and unsupervised methods, including regression, classification, PCA, NMF, resampling, and matched-filter techniques.
- Mentor students, coordinate project timelines, and promote reproducible and maintainable coding practices.

Postdoctoral Fellow *Lawrence Berkeley National Laboratory, USA* *Sep 2022 – Oct 2025*

Data Science and Software Development

- Developed and validated analytical models for large scientific datasets, including a hybrid PCA-based method that reduced major classification failures by approximately **30%**.
- Defined and tracked analysis-quality metrics, including failure rates, detection reliability, completeness, and validation statistics, to guide model and pipeline improvements.
- Contributed to open-source software through feature development, debugging, code review, automated testing, documentation, and versioned releases.
- Managed collaborative development through GitHub, including issue tracking, pull requests, CI/CD workflows, and tagged releases.

Large-Scale Computing and Project Leadership

- Supported daily production workflows on NERSC systems using Python, Slurm, parallel processing, and automated validation.
- Investigated workflow failures, identified root causes, and coordinated fixes across data, software, and computing teams.
- Led analytical and software workstreams from problem definition through implementation, testing, delivery, and communication of results.
- Mentored junior researchers and coordinated technical priorities, milestones, and project deliverables.

PhD Research Fellow *Max Planck Institute for Astrophysics, Germany* *Sep 2018 – Jul 2022*

- Developed a Python-based signal-detection tool for identifying weak features in noisy one-dimensional data.
- Built parallel analysis workflows that achieved over 95% detection reliability and reduced runtime from weeks to hours.
- Performed large-scale statistical validation using resampling, completeness testing, uncertainty analysis, and reproducible pipelines.
- Led projects from method design and implementation through validation, publication, and release of reusable data products.

Selected Open-Source Software

- **qsoabsfind** — Developer and maintainer of a scalable Python package for signal detection, classification, and measurement in noisy sequential data.
- **nmfqsofit** — Developer and maintainer of a Python package for data reconstruction and feature modeling using non-negative matrix factorization.
- **redrock** and **desispec** — Contributor to large-scale data-processing and model-fitting software used by an international scientific collaboration.

Technical Skills

Data Science & ML	Python, NumPy, SciPy, pandas, scikit-learn, regression, classification, clustering, PCA, NMF, model validation
Statistics & Signals	Statistical inference, probability, hypothesis testing, bootstrapping, uncertainty analysis, model comparison, observational-data validation, one-dimensional sequential data, convolution, matched filtering, feature detection, noise modeling
Data Engineering	Basic SQL for data access, data cleaning, data transformation, ETL-style workflows, structured and unstructured data, FITS, HDF5, parallel I/O
Software Development	Git, GitHub, code review, unit testing (*pytest*), GitHub Actions, CI/CD, automated validation, modular package design, Sphinx documentation
Large-Scale Computing	Linux, NERSC Perlmutter, Slurm, multiprocessing, MPI, performance debugging, Jupyter
Leadership	Project ownership, mentorship, cross-team collaboration, technical communication, project coordination

Selected Professional Activities

- Awarded Builder status, given to approximately 10% of collaboration members, for contributions to survey operations and data systems in one international project.
- Core member of the institute Computing Facilities Committee, contributing to research-computing planning and user support.
- Awarded a three-year Max Planck India Mobility Grant (2026–2029), given by Max Planck Society of Germany.
- Author of 48 publications, including 5 first-author papers, with more than 6000 citations.
- Referee for *Nature Astronomy*, *Astronomy & Astrophysics*, and *The Astrophysical Journal*.

Education

PhD in Astrophysics	*Max Planck Institute for Astrophysics, Garching, Germany*	*Sep 2018 – Jul 2022*
BS–MS in Physics	*Indian Institute of Science (IISc), Bangalore, India*	*Aug 2012 – Jun 2017*